

ENGR 213: Applied Ordinary Differential Equations

Paradis notes — In-Class Lecture, Tutorial & Midterm 1 Preparation Master Compendium

Concordia University · Gina Cody School of Engineering · Department of MIAE

Curriculum Authority: Dr. Alexandre Paradis, ing. Ph.D. · In-Class Lecture Notes 1–5 · Tutorials 1 & 3 · Homework Sets 1–3
Examination Target: Midterm #1 (October 19, 2026) · **Scope:** Chapter 2 (Sections 2.1–2.8) + Chapter 17 (Sections 17.1 & 17.2)

Department of Mechanical, Industrial & Aerospace Engineering · Concordia University

****Academic Session:**** Fall 2026 / Winter 2025 · ****Course Code:**** ENGR 213
****Instructor:**** Dr. Alexandre Paradis, ing. Ph.D.
****Curriculum Mapping:**** Lectures 1–5 · Tutorials 1 & 3 · Homework Sets 1–3 · Midterm #1 Examination Scope (October 19: Chapter 2 + §17.1 & §17.2)
This compendium provides an authoritative, rigorously structured, and comprehensive mathematical synthesis of Dr. Alexandre Paradis's in-class lectures, weekly tutorials, assigned textbook homework sets, and Midterm #1 preparation. All handwritten derivations and whiteboard notes from the official course documentation have been transformed into publication-grade mathematical typesetting with complete proofs, physical interpretations, verification steps, and high-yield examination strategies.

Curriculum Phase	Topics & Sections Covered	Primary Analytical Competencies
Module 1 (Lecture 1)	§1.1, §1.2: ODE vs. PDE, Order, Degree, Linearity	Classification criteria, Leibniz/Lagrange notations, Linearity Rules
Module 2 (Tutorial 1 & HW 1)	§1.1, §2.1: Solution Verification, Phase Line Stability	Radical domain restrictions, autonomous critical points, attractors vs. repellers
Module 3 (Lecture 2 & HW 2)	§2.2: Separation of Variables, Singular Solutions	Differential separation, lost equilibrium solutions, explicit IVP intervals
Module 4 (Lecture 3 & HW 3)	§2.3: First-Order Linear Equations & Integrating Factors	Standard form normalization, integrating factor $I(x) = e^{\int P(x)dx}$, transient terms
Module 5 (Tutorial 3 & Lecture 4)	§2.4, §2.7: Exact Equations & Physical Models	Test for exactness, potential function $\Psi(x, y)$, radioactive decay, Newton's cooling
Module 6 (Lecture 4)	§2.5, §2.8: Substitutions & Nonlinear Models	Bernoulli $u = y^{1-n}$, homogeneous $y = ux$, logistic growth, Torricelli's tank
Module 7 (Lecture 5)	§17.1, §17.2: Complex Variables & Polar Forms	Modulus, principal argument in radians, Euler's formula, De Moivre's theorem, n -th roots
Module 8 (Midterm 1 Strategy)	Midterm 1 Examination Synthesis & Pitfalls	10 high-yield exam traps, rapid check algorithms, calculator policy

Module 1: Fundamental Concepts & Classification (Lecture 1)

****Date Delivered:**** September 8, 2026 · ****Textbook Reference:**** Zill, Sections 1.1, 1.2 & 2.1

1.1 Course Administration & Policy

Approved Calculator Policy: Concordia University Gina Cody School of Engineering enforces a strict calculator policy. Only standard approved models bearing the official ENCS faculty sticker are permitted in examinations.**

Midterm #1 Date: Monday, October 19, 2026.**

Midterm #1 Scope: Chapter 2 (Sections 2.1–2.8) + Chapter 17 (Sections 17.1 & 17.2).**

1.2 Definition of Differential Equations

A differential equation is any mathematical relation involving one or more dependent variables (unknown functions) and one or more of their derivatives with respect to one or more independent variables.

Ordinary Differential Equation (ODE): Involves derivatives with respect to only a single independent variable (e.g., x or t):**

$$\frac{dy}{dx} + y = 0, \quad \left(\frac{dy}{dx}\right)^2 + 2xy = \cos(x), \quad \frac{d^2y}{dt^2} + 4y = 0$$

Partial Differential Equation (PDE): Involves partial derivatives of a multivariable function with respect to two or more independent variables:**

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad \frac{\partial y}{\partial t} = \alpha^2 \frac{\partial^2 y}{\partial x^2}$$

1.3 Standard Derivative Notations

Leibniz Notation: $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$, $\dots$, $\frac{d^ny}{dx^n}$ (essential for differential operations, separation of variables, and chain rule manipulations).**

Prime (Lagrange) Notation: y' , y'' , y''' , $y^{(4)}$, $\dots$, $y^{(n)}$ (compact and standard for linear constant-coefficient ODEs).**

Newton Dot Notation: $\dot{y} = \frac{dy}{dt}$, $\ddot{y} = \frac{d^2y}{dt^2}$ (standard in engineering dynamics, kinematics, and vibrations).**

1.4 Order and Degree of an ODE

Order: The order of an ODE is defined strictly as the order of the highest derivative present in the equation.**

Degree: The power (exponent) to which the highest derivative is raised, after the equation has been rationalized with respect to derivatives.**

$$\frac{dy}{dx} - y^4 = 0 \implies 1^{\text{st}} \text{ Order, Degree 1} \quad (\text{power of } y \text{ does not alter order})$$

$$\left(\frac{dy}{dx}\right)^3 + y = 0 \implies 1^{\text{st}} \text{ Order, Degree 3}$$

$$\frac{d^4y}{dx^4} + \frac{d^2y}{dx^2} + 1 = 0 \implies 4^{\text{th}} \text{ Order, Degree 1}$$

1.5 Linearity Criteria

An n -th order ODE is **linear if and only if it can be expressed in the standard canonical form:**

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

****The Two Golden Rules of Linearity:****

1. **Rule 1 (First Power Only):** The dependent variable y and every single one of its derivatives $y', y'', \dots, y^{(n)}$ appear to the first power only (no terms like $(y')^2$, y^3 , or $\sqrt{y}$).

2. **Rule 2 (Independent Coefficients):** The coefficient functions $a_n(x), \dots, a_0(x)$ and the non-homogeneous driving term $g(x)$ depend strictly and solely on the independent variable x (no cross-products such as yy' or transcendental functions of y like $\sin(y)$ or e^y).

Differential Equation	Order	Linear?	Mathematical Justification
$(y - x)dx + 4xdy = 0$	1	**Yes**	Dividing by dx : $4xy' + y = x$ satisfies standard linear form.
$y' = y^2 - 3y$	1	**No**	Nonlinear due to the quadratic power y^2 .
$y'' + 4y = \sin(x)$	2	**Yes**	Dependent variable y and y'' are linear; $\sin(x)$ depends only on x .
$y'' + 4 \sin(y) = 0$	2	**No**	Nonlinear due to transcendental function $\sin(y)$ of the dependent variable.
$x^2y''' + xy' - 5y = e^x$	3	**Yes**	3rd-order linear with variable coefficients $x^2, x, -5$.

Module 2: In-Class Tutorial 1 & Homework 1 Solutions

****Date Delivered:** September 14, 2026 • **Focus:** Solution Verification, Radical Domains & Autonomous Phase Portraits**

2.1 Problem §1.1 Q13: First-Order Linear Verification

Given ODE & Trial Solution:**

$$2y' + y = 0, \quad y(x) = e^{-x/2}$$

Step-by-Step Analytical Derivation:**

1. Compute the first derivative using the exponential chain rule:

$$y'(x) = \frac{d}{dx} [e^{-x/2}] = -\frac{1}{2}e^{-x/2}$$

2. Substitute $y(x)$ and $y'(x)$ into the left-hand differential operator:

$$\text{LHS} = 2y' + y = 2 \left(-\frac{1}{2}e^{-x/2} \right) + e^{-x/2} = -e^{-x/2} + e^{-x/2} = 0$$

3. Since LHS = 0 = RHS, the function satisfies the ODE identically for all $x \in \mathbb{R}$.

Interval of Definition:**

$$I = (-\infty, +\infty)$$

2.2 Problem §1.1 Q17: Radical Solution & Domain Restriction

Given ODE & Trial Solution:**

$$(y - x)y' = y - x + 8, \quad y(x) = x + 4\sqrt{x+2}$$

Step-by-Step Analytical Derivation:**

1. Differentiate the trial function:

$$y'(x) = \frac{d}{dx} [x + 4(x+2)^{1/2}] = 1 + 4 \left(\frac{1}{2\sqrt{x+2}} \right) = 1 + \frac{2}{\sqrt{x+2}}$$

2. Evaluate the Left-Hand Side (LHS):

$$y - x = (x + 4\sqrt{x+2}) - x = 4\sqrt{x+2}$$

$$\text{LHS} = (y - x)y' = (4\sqrt{x+2}) \left(1 + \frac{2}{\sqrt{x+2}} \right) = 4\sqrt{x+2} + \frac{8\sqrt{x+2}}{\sqrt{x+2}} = 4\sqrt{x+2} + 8$$

3. Evaluate the Right-Hand Side (RHS):

$$\text{RHS} = y - x + 8 = (x + 4\sqrt{x+2}) - x + 8 = 4\sqrt{x+2} + 8$$

4. Notice LHS $\equiv$ RHS identically.

Interval of Validity Determination:**

The square root $\sqrt{x+2}$ requires $x+2 \geq 0 \implies x \geq -2$. However, in the derivative

$y'(x) = 1 + \frac{2}{\sqrt{x+2}}$, the term $\sqrt{x+2}$ appears in the denominator, requiring $x+2 \neq 0 \implies x > -2$.

$$I = (-2, +\infty)$$

2.3 Problem §1.1 Q25: Second-Order Homogeneous Verification

Given ODE & Trial Solution:**

$$y'' - 4y' + 4y = 0, \quad y(x) = c_1 e^{2x} + c_2 x e^{2x}$$

Step-by-Step Analytical Derivation:**

1. Write the trial function in factored form: $y(x) = (c_1 + c_2 x)e^{2x}$.

2. Compute the first derivative via product rule:

$$y'(x) = c_2 e^{2x} + 2(c_1 + c_2 x)e^{2x} = (2c_1 + c_2 + 2c_2 x)e^{2x}$$

3. Compute the second derivative:

$$y''(x) = 2c_2 e^{2x} + 2(2c_1 + c_2 + 2c_2 x)e^{2x} = (4c_1 + 4c_2 + 4c_2 x)e^{2x}$$

4. Substitute into the differential operator:

$$\begin{aligned} y'' - 4y' + 4y &= e^{2x} [(4c_1 + 4c_2 + 4c_2 x) - 4(2c_1 + c_2 + 2c_2 x) + 4(c_1 + c_2 x)] \\ &= e^{2x} [(4c_1 - 8c_1 + 4c_1) + (4c_2 - 4c_2) + (4c_2 x - 8c_2 x + 4c_2 x)] = e^{2x} [0] \equiv 0 \end{aligned}$$

Interval of Definition:** $I = (-\infty, +\infty)$.

2.4 Problem §2.1 Q21: Autonomous Direction Fields & Stability

Given Autonomous ODE:**

$$\frac{dy}{dx} = y^2 - 3y = y(y - 3)$$

Critical Points (Equilibrium Solutions):**

Set $\frac{dy}{dx} = 0 \implies y(y - 3) = 0 \implies y_1 = 0 \quad \text{and} \quad y_2 = 3$.

Phase Line Sign Analysis:**

****Interval $(-\infty, 0)$:** Test $y = -1 \implies (-1)(-4) = +4 > 0 \implies \frac{dy}{dx} > 0$ (arrows point UP toward $y = 0$).**

****Interval $(0, 3)$:** Test $y = 1 \implies (1)(-2) = -2 < 0 \implies \frac{dy}{dx} < 0$ (arrows point DOWN toward $y = 0$).**

****Interval $(3, +\infty)$:** Test $y = 4 \implies (4)(1) = +4 > 0 \implies \frac{dy}{dx} > 0$ (arrows point UP away from $y = 3$).**

Stability Classification:**

**** $y = 0$ is Asymptotically Stable (Attractor / Sink):**** Nearby solution trajectories converge to 0 as $x \rightarrow +\infty$.

**** $y = 3$ is Unstable (Repeller / Source):**** Nearby solution trajectories diverge away from 3 as $x \rightarrow +\infty$.

Module 3: Separable Differential Equations & IVPs (Lectures 2 & 3)

****Dates Delivered:** September 10 & 22, 2026 • **Textbook Reference:** Zill, Section 2.2**

3.1 Mathematical Theory of Separable Equations

A first-order differential equation is **separable** if the right-hand side can be factored into a product of a function of x and a function of y :

$$\frac{dy}{dx} = g(x)f(y)$$

Standard 4-Step Analytical Algorithm:

1. **Check for Singular Solutions:** Set $f(y) = 0$. Any constant root $y = c$ is an equilibrium solution. Check at the end whether it is included in the general family.

2. **Separate Differentials:** Divide by $f(y)$ and multiply by dx :

$$\frac{1}{f(y)} dy = g(x) dx \quad (f(y) \neq 0)$$

3. **Integrate Both Sides:**

$$\int \frac{1}{f(y)} dy = \int g(x) dx + C$$

4. **Isolate $y(x)$ and Apply Initial Conditions:** Use initial condition $y(x_0) = y_0$ to isolate C and explicitly define the solution's interval of definition.

3.2 Dr. Paradis In-Class Quiz 1 Master Problem

Problem Statement: Solve the initial value problem:**

$$y \frac{dy}{dx} + x^2 = 4, \quad y(0) = 2$$

Complete Step-by-Step Derivation:**

1. Isolate the derivative term:

$$y \frac{dy}{dx} = 4 - x^2$$

2. Separate variables:

$$y \, dy = (4 - x^2) \, dx$$

3. Integrate both sides:

$$\int y \, dy = \int (4 - x^2) \, dx \implies \frac{y^2}{2} = 4x - \frac{x^3}{3} + C_1$$

4. Multiply by 2:

$$y^2 = 8x - \frac{2}{3}x^3 + C$$

5. Apply initial condition $y(0) = 2$:

$$(2)^2 = 8(0) - \frac{2}{3}(0)^3 + C \implies 4 = C$$

6. Substitute $C = 4$:

$$y^2 = 8x - \frac{2}{3}x^3 + 4$$

7. Since the initial condition specifies $y(0) = +2 > 0$, we select the positive branch:

$$y(x) = \sqrt{8x - \frac{2}{3}x^3 + 4}$$

Interval of Definition:**

The solution is valid on the open interval containing $x = 0$ where the radicand remains strictly positive: $8x - \frac{2}{3}x^3 + 4 > 0$.

3.3 Problem §2.2 Q17: Algebraic Factorization Separable ODE

Given ODE:**

$$\frac{dy}{dx} = \frac{xy + 3x - y - 3}{xy - 2x + 4y - 8}$$

Analytical Derivation:**

1. Factor numerator and denominator by grouping:

$$\text{Numerator} = x(y + 3) - (y + 3) = (x - 1)(y + 3)$$

$$\text{Denominator} = x(y - 2) + 4(y - 2) = (x + 4)(y - 2)$$

2. Rewrite the separated equation:

$$\frac{y - 2}{y + 3} dy = \frac{x - 1}{x + 4} dx$$

3. Perform polynomial division on both sides:

$$\frac{y - 2}{y + 3} = \frac{(y + 3) - 5}{y + 3} = 1 - \frac{5}{y + 3}$$

$$\frac{x - 1}{x + 4} = \frac{(x + 4) - 5}{x + 4} = 1 - \frac{5}{x + 4}$$

4. Integrate both sides:

$$\int \left(1 - \frac{5}{y + 3}\right) dy = \int \left(1 - \frac{5}{x + 4}\right) dx$$

$$y - 5 \ln |y + 3| = x - 5 \ln |x + 4| + C$$

5. Rearrange terms:

$$y - x + C = 5 \ln \frac{y + 3}{x + 4}$$

Module 4: First-Order Linear Equations & Integrating Factors (Lecture 3 & HW 3)

****Dates Delivered:** September 22 & 25, 2026 • ****Textbook Reference:** Zill, Section 2.3****

4.1 Standard Canonical Form

A first-order linear differential equation must always be written with the leading coefficient equal to +1:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

4.2 The Integrating Factor Theorem

Multiply the entire equation by an unknown strictly positive integrating factor $I(x)$:

$$I(x) \frac{dy}{dx} + I(x)P(x)y = I(x)Q(x)$$

We demand that the left-hand side matches the derivative of the product $[I(x)y]$:

$$\frac{d}{dx}[I(x)y] = I(x) \frac{dy}{dx} + I'(x)y$$

Equating coefficients of y :

$$I'(x) = I(x)P(x) \implies \frac{dI}{I} = P(x) dx \implies \ln |I| = \int P(x) dx \implies I(x) = e^{\int P(x) dx}$$

4.3 General Solution Formula

Integrating both sides of $\frac{d}{dx}[I(x)y] = I(x)Q(x)$:

$$I(x)y = \int I(x)Q(x) dx + C \implies y(x) = \frac{1}{I(x)} \left[\int I(x)Q(x) dx + C \right]$$

IMPORTANT EXAM RULE: **Dr. Paradis Exam Golden Rule: The arbitrary constant $+C$ MUST be placed inside the brackets before dividing by $I(x)$. Writing $y = \frac{1}{I(x)} \int I(x)Q(x) dx + C$ is an automatic deduction of full method marks!**

4.4 Transient Terms vs. Steady-State Solutions

The general solution can be partitioned into:

$$y(x) = y_c(x) + y_p(x) = \frac{C}{I(x)} + \frac{1}{I(x)} \int I(x)Q(x) dx$$

Transient Term: Any component of $y(x)$ that decays to zero as $x \rightarrow +\infty$ (typically terms with e^{-kx} for $k > 0$):**

$$\lim_{x \rightarrow \infty} y_{\text{transient}}(x) = 0$$

Steady-State Term: The persistent long-term behavior of the system as $x \rightarrow +\infty$.**

4.5 Problem §2.3 Q21: Singular Points and Interval of Definition

Given Equation:**

$$x \frac{dy}{dx} + 2y = 3, \quad x > 0$$

Step-by-Step Derivation:**

1. Divide by x to achieve standard form:

$$\frac{dy}{dx} + \frac{2}{x}y = \frac{3}{x} \implies P(x) = \frac{2}{x}, \quad Q(x) = \frac{3}{x}$$

2. Compute integrating factor:

$$I(x) = \exp\left(\int \frac{2}{x} dx\right) = e^{2 \ln|x|} = e^{\ln(x^2)} = x^2$$

3. Apply product rule:

$$\frac{d}{dx}[x^2 y] = x^2 \left(\frac{3}{x}\right) = 3x$$

4. Integrate both sides:

$$x^2 y = \int 3x dx + C = \frac{3}{2}x^2 + C \implies y(x) = \frac{3}{2} + \frac{C}{x^2}$$

5. The term $\frac{C}{x^2}$ is the **transient term since $\lim_{x \rightarrow \infty} \frac{C}{x^2} = 0$. The **steady-state solution** is $y_{ss} = \frac{3}{2}$.**

6. The point $x = 0$ is a **singular point where $P(x)$ and $Q(x)$ are discontinuous. Thus, the solution exists on $I = (0, +\infty)$.**

Module 5: Exact Equations & Physical Modeling (Tutorial 3 & Lecture 4)

****Dates Delivered:** September 28 & 29, 2026 • **Textbook Reference:** Zill, Sections 2.4, 2.7 & 2.8**

5.1 Test for Exactness (§2.4)

A first-order differential expression in differential form:

$$M(x, y) dx + N(x, y) dy = 0$$

is an **exact differential in a simply connected region R if and only if:**

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

5.2 Construction of Potential Function $\Psi(x, y)$

When exactness holds, there exists a potential function $\Psi(x, y)$ such that $\frac{\partial \Psi}{\partial x} = M$ and $\frac{\partial \Psi}{\partial y} = N$.

1. Integrate $M(x, y)$ with respect to x , introducing an arbitrary function of integration $g(y)$:

$$\Psi(x, y) = \int M(x, y) dx + g(y)$$

2. Differentiate $\Psi(x, y)$ with respect to y and set equal to $N(x, y)$:

$$\frac{\partial \Psi}{\partial y} = \frac{\partial}{\partial y} \left[\int M(x, y) dx \right] + g'(y) = N(x, y)$$

3. Isolate $g'(y)$ (verify all x variables cancel out) and integrate to find $g(y)$.

4. The implicit solution is:

$$\Psi(x, y) = C$$

5.3 Special Integrating Factors for Non-Exact Equations

If $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, compute the test ratios:

Function of x alone:**

$$\frac{M_y - N_x}{N} = f(x) \implies \mu(x) = e^{\int f(x) dx}$$

Function of y alone:**

$$\frac{N_x - M_y}{M} = g(y) \implies \mu(y) = e^{\int g(y) dy}$$

5.4 Tutorial 3 Worked Problem 1: Radioactive Decay (\$2.7)

Problem Statement:** An initial radioactive mass $A(0) = 100$ mg decreases by 3% after 6 hours. Find the remaining mass at $t = 24$ hours.

Analytical Derivation:**

1. Decay model: $\frac{dA}{dt} = -kA \implies A(t) = A_0 e^{-kt} = 100e^{-kt}$.

2. At $t = 6$ h, 3% has decayed, so 97% remains:

$$A(6) = 100e^{-6k} = 97 \implies e^{-6k} = 0.97 \implies -k = \frac{1}{6} \ln(0.97)$$

3. At $t = 24 = 4 \times 6$ hours:

$$A(24) = 100e^{-24k} = 100(e^{-6k})^4 = 100(0.97)^4 \approx 88.53 \text{ mg}$$

5.5 Tutorial 3 Worked Problem 2: Forensic Newton's Law of Cooling (§2.7)

Problem Statement: A body is discovered in a room maintained at constant ambient temperature $T_m = 70^\circ\text{F}$. At time of discovery t_d , the body temperature is 85°F . One hour later ($t_d + 1$), the temperature drops to 80°F . Assuming living body temperature was $T(0) = 98.6^\circ\text{F}$, estimate the time of death.**

Analytical Derivation:**

1. Newton's Cooling ODE:

$$\frac{dT}{dt} = -k(T - 70) \implies \int \frac{dT}{T - 70} = -k \int dt \implies T(t) = 70 + Ce^{-kt}$$

2. At time of death $t = 0$:

$$T(0) = 70 + C = 98.6 \implies C = 28.6 \implies T(t) = 70 + 28.6e^{-kt}$$

3. At discovery t_d :

$$85 = 70 + 28.6e^{-kt_d} \implies 28.6e^{-kt_d} = 15 \implies e^{-kt_d} = \frac{15}{28.6}$$

4. One hour later at $t_d + 1$:

$$80 = 70 + 28.6e^{-k(t_d+1)} \implies 28.6e^{-kt_d}e^{-k} = 10$$

5. Substitute $28.6e^{-kt_d} = 15$:

$$15e^{-k} = 10 \implies e^{-k} = \frac{10}{15} = \frac{2}{3} \implies k = -\ln\left(\frac{2}{3}\right) \approx 0.4055 \text{ hr}^{-1}$$

6. Calculate elapsed time t_d :

$$-kt_d = \ln\left(\frac{15}{28.6}\right) \implies t_d = \frac{\ln(15/28.6)}{\ln(2/3)} = \frac{-0.6453}{-0.4055} \approx 1.59 \text{ hours (1 hr 35 min)}$$

Conclusion: Death occurred approximately 1 hour and 35 minutes prior to discovery.**

Module 6: Solutions by Substitution & Nonlinear Models (Lectures 4 & 5)

****Dates Delivered:** September 29 & October 1, 2026 • **Textbook Reference:** Zill, Sections 2.5 & 2.8**

6.1 Bernoulli Differential Equations

A first-order equation of the form:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \quad (n \neq 0, 1)$$

is transformed into a linear ODE via the substitution:

$$u = y^{1-n} \implies \frac{du}{dx} = (1-n)y^{-n} \frac{dy}{dx}$$

Dividing the original ODE by y^n :

$$y^{-n} \frac{dy}{dx} + P(x)y^{1-n} = Q(x) \implies \frac{du}{dx} + (1-n)P(x)u = (1-n)Q(x)$$

6.2 Homogeneous Differential Equations of Degree Zero

An ODE $M(x, y)dx + N(x, y)dy = 0$ where M and N are homogeneous functions of the exact same degree can be written as:

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

Substitute:

$$y = ux \implies \frac{dy}{dx} = u + x \frac{du}{dx}$$

Substituting into the ODE yields an immediately separable equation:

$$u + x \frac{du}{dx} = f(u) \implies \frac{du}{f(u) - u} = \frac{dx}{x}$$

6.3 Linear Arguments: $y' = f(Ax + By + C)$

When the derivative depends strictly on a linear combination $Ax + By + C$ ($B \neq 0$), substitute:

$$u = Ax + By + C \implies \frac{du}{dx} = A + B \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{B} \left(\frac{du}{dx} - A \right)$$

Separated form:

$$\frac{du}{A + Bf(u)} = dx$$

6.4 Nonlinear Physical Models (§2.8)

Logistic Population Growth:**

$$\frac{dP}{dt} = P(a - bP) = aP \left(1 - \frac{P}{K}\right), \quad K = \frac{a}{b} \text{ (Carrying Capacity)}$$

Analytical solution via partial fractions or Bernoulli substitution ($n = 2$):

$$P(t) = \frac{K}{1 + \left(\frac{K-P_0}{P_0}\right)e^{-at}}$$

Torricelli's Law (Draining Tank):**

$$\frac{dV}{dt} = -A_h v_{\text{exit}} = -A_h \sqrt{2gh}$$

For a cylindrical container with constant top surface area A_w :

$$A_w \frac{dh}{dt} = -A_h \sqrt{2gh} \implies \frac{dh}{\sqrt{h}} = -\frac{A_h \sqrt{2g}}{A_w} dt$$

Module 7: Complex Numbers & Polar Coordinates (Lectures 4 & 5)

****Dates Delivered:** September 29 & October 1, 2026 • ****Textbook Reference:** Zill, Chapter 17 (Sections 17.1 & 17.2)****

7.1 Cartesian Representation (§17.1)

A complex number $z \in \mathbb{C}$ is defined as an ordered pair of real numbers (x, y) :

$$z = x + iy \quad (i = \sqrt{-1}, i^2 = -1)$$

Real Part: $\text{Re}(z) = x$**

Imaginary Part: $\text{Im}(z) = y$ (note: $\text{Im}(z)$ is the real coefficient y , NOT iy)**

Complex Conjugate: $\bar{z} = x - iy$**

Fundamental Modulus Identity:**

$$z\bar{z} = (x + iy)(x - iy) = x^2 - (iy)^2 = x^2 + y^2 = |z|^2$$

Division by Conjugate Rationalization:**

$$\frac{z_1}{z_2} = \frac{z_1 \bar{z}_2}{z_2 \bar{z}_2} = \frac{(x_1 + iy_1)(x_2 - iy_2)}{x_2^2 + y_2^2} = \frac{(x_1 x_2 + y_1 y_2) + i(x_2 y_1 - x_1 y_2)}{x_2^2 + y_2^2}$$

7.2 Polar & Exponential Representation (§17.2)

Modulus (Length):**

$$r = |z| = \sqrt{x^2 + y^2}$$

Argument (Phase Angle):**

$$\theta = \arg(z) \implies \tan \theta = \frac{y}{x}$$

EXAM WARNING: **Dr. Paradis Golden Rule on Angles: All complex arguments in ENGR 213 MUST be computed in **radians**, never degrees ($\pi \text{ rad} = 180^\circ$).**

Principal Argument $\arg(z) = \Theta \in (-\pi, \pi]$:

$$\Theta = \begin{cases} \arctan(y/x), & x > 0 \\ \arctan(y/x) + \pi, & x < 0, y \geq 0 \\ \arctan(y/x) - \pi, & x < 0, y < 0 \\ \pi/2, & x = 0, y > 0 \\ -\pi/2, & x = 0, y < 0 \end{cases}$$

Euler's Formula & Polar Form:**

$$e^{i\theta} = \cos \theta + i \sin \theta \implies \mathbf{z} = \mathbf{r}(\cos \theta + \mathbf{i} \sin \theta) = \mathbf{r}e^{i\theta}$$

7.3 Arithmetic Operations in Polar Coordinates

Let $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$:

1. **Multiplication:**

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)} = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

(Moduli multiply; arguments add)

2. **Division:**

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$$

(Moduli divide; arguments subtract)

3. **De Moivre's Theorem:**

$$z^n = (r e^{i\theta})^n = r^n e^{in\theta} = \mathbf{r}^n [\cos(n\theta) + \mathbf{i} \sin(n\theta)]$$

4. ** n -th Roots of Complex Numbers:**

The n distinct roots $w_k = z^{1/n}$ of $z = r e^{i\theta}$ are given by:

$$\mathbf{w}_k = \sqrt[n]{\mathbf{r}} \exp\left(\mathbf{i} \frac{\theta + 2\mathbf{k}\pi}{\mathbf{n}}\right) = \sqrt[n]{\mathbf{r}} \left[\cos\left(\frac{\theta + 2\mathbf{k}\pi}{\mathbf{n}}\right) + \mathbf{i} \sin\left(\frac{\theta + 2\mathbf{k}\pi}{\mathbf{n}}\right) \right]$$

for $k = 0, 1, 2, \dots, n-1$.

Module 8: Midterm #1 Strategic Preparation & Pitfall Elimination (October 19)

8.1 Ten High-Yield Examination Traps & Checklists

1. ****Trap 1: Coefficient Normalization in Linear ODEs:**** Never calculate $P(x)$ until the coefficient of $\frac{dy}{dx}$ has been divided out to equal +1.
 2. ****Trap 2: Bracketing the Integrating Factor:**** The arbitrary constant $+C$ belongs strictly inside the brackets: $y = \frac{1}{I(x)} \left[\int IQdx + C \right]$.
 3. ****Trap 3: Absolute Values in Logarithms:**** $\int \frac{1}{x} dx = \ln|x| \implies e^{\ln|x|} = |x|$. Specify whether $x > 0$ or $x < 0$ to drop the absolute value cleanly.
 4. ****Trap 4: Lost Singular Solutions:**** When separating $\frac{dy}{f(y)} = g(x)dx$, immediately check if $f(y) = 0$ yields constant solutions that are not covered by any value of C .
 5. ****Trap 5: Exactness Cross-Derivative Signs:**** Exactness requires $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. Do not confuse the partial derivative order.
 6. ****Trap 6: Correct Bernoulli Exponent:**** The substitution is $u = y^{1-n}$, which transforms the equation into $\frac{du}{dx} + (1-n)P(x)u = (1-n)Q(x)$.
 7. ****Trap 7: Radians vs. Degrees in Complex Arithmetic:**** Dr. Paradis penalizes degrees on complex polar exam problems. Ensure calculator mode is in ****radians****.
 8. ****Trap 8: Complex Roots Spacing:**** When finding n -th roots, always add $\frac{2k\pi}{n}$ for $k = 0, 1, \dots, n-1$ to capture all n distinct roots evenly distributed on the circle.
 9. ****Trap 9: Open Intervals of Definition:**** Initial value problem solutions must be stated on open intervals ($I = (-2, \infty)$ rather than $[-2, \infty)$) where the derivative is defined.
 10. ****Trap 10: Sign Conventions in Decay & Cooling:**** Rate constant $k > 0$ means decay is written as $\frac{dA}{dt} = -kA$. Cooling is $\frac{dT}{dt} = -k(T - T_m)$.
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